

9

Fiber Polytopes, and Beyond

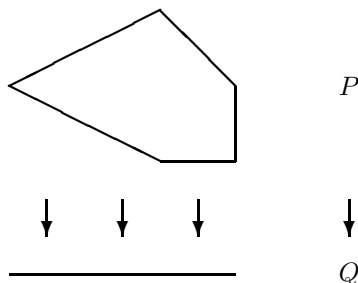
The *secondary polytopes* of Gel'fand, Kapranov & Zelevinsky [231] are a quite recent development that grew out of their theory of “ $\mathcal{A}$ -hypergeometric functions.” After Izrail M. Gel'fand and Andrei Zelevinsky presented the miraculous construction at the 1989 Symposium on Combinatorics and Geometry in Stockholm, a lot of effort was put into understanding what is going on geometrically. It seems that the definitive answer — unexpectedly simple — is the one supplied by Louis Billera and Bernd Sturmfels [78, 79, 534], who introduced the *fiber polytope* of a projection of polytopes and showed that secondary polytopes arise in the special case where we project the $(n - 1)$ -simplex to a given polytope with n vertices.

The main goal in this lecture is to develop geometric intuition for the fiber polytope construction. Many interesting examples have been studied, and among them we will concentrate on the construction of the permutahedron and the associahedron as fiber polytopes. With “fiber polytopes intuition” we then construct the permuto-associahedra: nice new polytopes proposed by Kapranov [313] as combinatorial objects and realized as polytopes in Reiner & Ziegler [453].

So fiber polytopes help to solve special cases of a difficult general problem: the *construction of polytopes with specified combinatorics*. Using the Lawrence construction (Lecture 6), one can see that the solution for any specified face lattice is difficult: this “algorithmic Steinitz problem” is as difficult as the solution of general polynomial systems over the reals; see [96, p. 407]. (The inverse problem, the complete description of the face lattice of a given polytope, is not trivial either: see Lecture 1 and Exercise 9.0).

9.1 Polyhedral Subdivisions and Fiber Polytopes

The basic object of study is a *projection* of polytopes $\pi : P \longrightarrow Q$, that is, an affine map $\pi : \mathbb{R}^p \longrightarrow \mathbb{R}^q$ such that $\pi(P) = Q$, for polytopes $P \subseteq \mathbb{R}^p$ and $Q \subseteq \mathbb{R}^q$. We may assume that P is a p -dimensional polytope, and Q is a q -polytope. A simple example, for $p = 2$, $q = 1$, is drawn here.



Definition 9.1. Let $\pi : \mathbb{R}^p \longrightarrow \mathbb{R}^q$, $\pi(P) = Q$ be a projection of polytopes.

A π -induced subdivision $\pi(\mathcal{F})$ of Q is a polyhedral complex that subdivides Q , with the following two conditions:

- (i) The subdivision is of the form $\{\pi(F) : F \in \mathcal{F}\}$, for some specified collection $\mathcal{F} \subseteq L(P)$ of faces of P .
- (ii) $\pi(F) \subseteq \pi(F')$ implies $F = F' \cap \pi^{-1}(\pi(F))$, and thus, in particular, $F \subseteq F'$.

Every polytope in a π -induced subdivision $\pi(\mathcal{F})$ arises from a unique face $F \in \mathcal{F}$, and the collection $\mathcal{F}$ is part of the definition of $\pi(\mathcal{F})$. Thus, we usually abuse notation and call the family of polytopes $\mathcal{F} \subseteq L(P)$ itself the π -induced subdivision.

We define a partial order on these subdivisions by setting

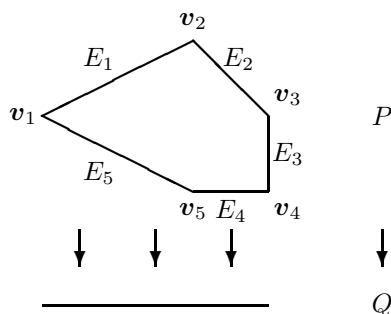
$$\mathcal{F}_1 \leq \mathcal{F}_2 \quad \text{if and only if} \quad \bigcup \mathcal{F}_1 \subseteq \bigcup \mathcal{F}_2.$$

So, $\mathcal{F}_1$ is “smaller” than $\mathcal{F}_2$ if the union of the polytopes in $\mathcal{F}_1$ is contained in the union of the polytopes in $\mathcal{F}_2$. This means that the subdivision $\{\pi(F) : F \in \mathcal{F}_1\}$ of Q is a *refinement* of the subdivision induced by $\mathcal{F}_2$.

The resulting partially ordered set, containing all subdivisions of Q that are induced by $\pi : P \longrightarrow Q$, will be denoted by

$$\omega(P, Q).$$

For example, consider the projection sketched before, and label the vertices and edges of P :



In this situation, there are three subdivisions of $Q \subseteq \mathbb{R}$ induced by the projection from $P \subseteq \mathbb{R}^2$, given by

$$\begin{aligned}\mathcal{F}_0 &= \{v_1, P, E_3\}, \\ \mathcal{F}_1 &= \{v_1, E_1, v_2, E_2, v_3\}, \\ \mathcal{F}_2 &= \{v_1, E_5, v_5, E_4, v_4\}.\end{aligned}$$

Note that the actual tilings $\pi(\mathcal{F}_1), \pi(\mathcal{F}_2)$ of Q coincide, but they are distinguished since they correspond to distinct collections $\mathcal{F}_1, \mathcal{F}_2 \subseteq L(P)$.

Condition (ii) in Definition 9.1 excludes “noncontinuous” sections like $\{v_1, E_1, v_2, E_4, v_4\}$. Thus, a π -induced subdivision of Q is given by a family $\mathcal{F}$ of faces of P — note that the projected set $\{\pi(F) : F \in \mathcal{F}\}$ is not sufficient to determine the subdivision as defined in Definition 9.1.

Condition (ii) is stronger than just requiring that $\pi(F) \subseteq \pi(F')$ implies $F \subseteq F'$: for example, families like $\{v_1, P, v_3\}$ are also excluded. Condition (ii) also implies that every collection $\mathcal{F} \subseteq L(P)$ is completely determined by its inclusion-maximal members.

The partial order that we get for our example is

$$\omega(P, Q) : \begin{array}{ccc} & \mathcal{F}_0 & \\ & \wedge & \\ \mathcal{F}_1 & & \mathcal{F}_2 \end{array}$$

which corresponds to the nonempty faces of a 1-dimensional polytope: we will see why ahead.

Note that $\dim(P) \geq \dim(Q)$ holds for every surjective map $\pi : P \rightarrow Q$ of polytopes. Thus, if $\mathcal{F} \subseteq L(P)$ describes a π -induced subdivision of Q , then we necessarily have $\dim(F) \geq \dim(\pi(F))$ for all $F \in \mathcal{F}$. If equality $\dim(F) = \dim(\pi(F))$ holds for all $F \in \mathcal{F}$, then the subdivision is called *tight*. The condition is equivalent to requiring $\dim(F) = \dim(\pi(F)) = q$ for all inclusion-maximal faces $F \in \mathcal{F}$. So, in the above example $\mathcal{F}_1$ and $\mathcal{F}_2$ are tight, but $\mathcal{F}_0$ is not.

We leave it as an exercise to prove that the tight subdivisions exactly correspond to the minimal elements of $\omega(P, Q)$, that is, to the minimal elements of the poset of π -induced subdivisions. (See Exercise 9.3, and also Lemma 9.5).

Recall from Section 5.1 that a polyhedral subdivision of Q is *regular* if it arises from all the “lower faces” of a polytope $\widehat{Q} \subseteq \mathbb{R}^q \times \mathbb{R}$ under the projection map $\pi : \mathbb{R}^q \times \mathbb{R} \rightarrow \mathbb{R}^q$ that forgets the last coordinate. Formally, the *lower faces* of $\widehat{Q}$ are those faces that minimize some linear function $(\mathbf{c}, c_0) \in (\mathbb{R}^q \times \mathbb{R})^*$ with $c_0 > 0$ over $\widehat{Q}$.

For example, in the preceding sketch the faces $\mathbf{v}_1$, E_5 , $\mathbf{v}_5$, E_4 , and $\mathbf{v}_4$ are the lower faces, and taking $\widehat{Q} := P$ defines the regular subdivision π -induced by $\mathcal{F}_2$.

The following construction yields regular π -induced subdivisions. (A different one is described in [79].)

Definition 9.2. Let $\pi : P \rightarrow Q$ be a projection of polytopes, and let $\mathbf{c} \in (\mathbb{R}^p)^*$. Then $\pi^{\mathbf{c}} : \mathbf{x} \mapsto \begin{pmatrix} \pi(\mathbf{x}) \\ \mathbf{c}\mathbf{x} \end{pmatrix}$ is a linear map from $\mathbb{R}^p$ to $\mathbb{R}^q \times \mathbb{R}$, so $\mathbf{c}$ determines a polytope

$$P \xrightarrow{\pi^{\mathbf{c}}} Q^{\mathbf{c}} := \left\{ \begin{pmatrix} \pi(\mathbf{x}) \\ \mathbf{c}\mathbf{x} \end{pmatrix} : \mathbf{x} \in P \right\} \subseteq \mathbb{R}^{q+1}$$

which projects down to Q via the map ρ that deletes the last coordinate. Let $\mathcal{L}^\downarrow(Q^{\mathbf{c}}) \subseteq L(Q^{\mathbf{c}})$ be the family of lower faces of $Q^{\mathbf{c}}$. Then

$$\mathcal{F}^{\mathbf{c}} := (\pi^{\mathbf{c}})^{-1} \mathcal{L}^\downarrow(Q^{\mathbf{c}}) = \{P \cap (\pi^{\mathbf{c}})^{-1}(F) : F \in \mathcal{L}^\downarrow(Q^{\mathbf{c}})\} \subseteq L(P)$$

induces a subdivision of Q . Such subdivisions will be called *π -coherent*. By

$$\omega_{coh}(P, Q)$$

we will denote the subposet of π -coherent subdivisions of Q , in the (usually larger) poset of all π -induced subdivisions.

Note that the subdivisions $\{\pi(F) : F \in \mathcal{F}\}$ that arise this way are regular by construction. Also π -coherent subdivisions are π -induced, because $\pi = \rho \circ \pi^{\mathbf{c}} : P \rightarrow Q$. It is not true that all the regular π -induced subdivisions are π -coherent, as we will see now. However, this is true if P is a simplex (see Exercise 9.5).

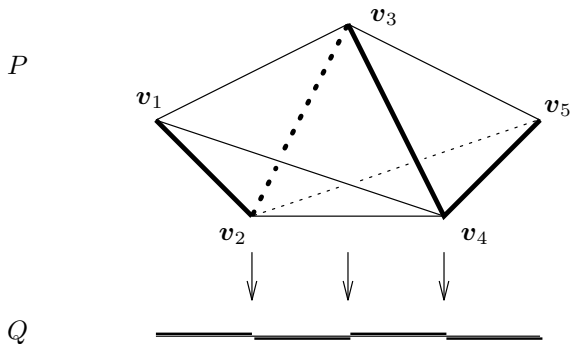
Example 9.3. Let $P = \text{conv}(V) \cong \text{bipyr}(\Delta_2)$ be the bipyramid over a triangle given by

$$V = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 2 & 0 & 1 \end{pmatrix},$$

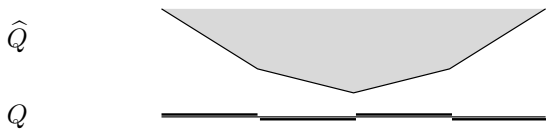
and let $\pi : P \rightarrow Q := [0, 4]$ be the projection to the first coordinate. Then the tight subdivision

$$\mathcal{F} = \{\mathbf{v}_1, [\mathbf{v}_1, \mathbf{v}_2], \mathbf{v}_2, [\mathbf{v}_2, \mathbf{v}_3], \mathbf{v}_3, [\mathbf{v}_3, \mathbf{v}_4], \mathbf{v}_4, [\mathbf{v}_4, \mathbf{v}_5], \mathbf{v}_5\}$$

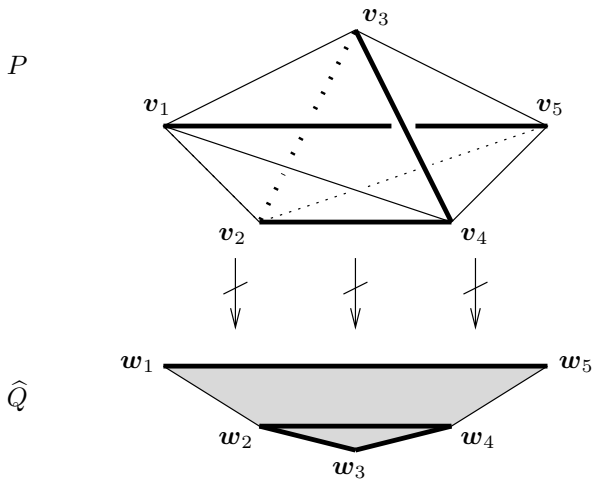
illustrated in the drawing is π -induced,



and it is *regular* (all subdivisions of a 1-dimensional polytope are regular).



However, this subdivision of Q is *not* π -coherent. In fact, if $\widehat{Q}$ is any such polytope whose lower faces yield the subdivision of Q , then there cannot be a linear map $\pi^{\mathbf{c}} : P \longrightarrow Q^{\mathbf{c}} \cong \widehat{Q}$ as required by Definition 9.2. In fact, such a map would have to take $\mathbf{v}_i$ to $\mathbf{w}_i$ in the following sketch.



However, this is impossible for an affine map, since the line $[\mathbf{v}_1, \mathbf{v}_5]$ intersects the triangle $[\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4]$ in P , and there is no such intersection of the triangle $[\mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4]$ with the line $[\mathbf{w}_1, \mathbf{w}_5]$ in $\hat{Q}$.

For a general projection of polytopes, the structure of the poset of *all* subdivisions is not clear. The “generalized Baues problem” of Billera, Kapranov & Sturmfels [73] asked whether this poset always has the “homotopy type of a $(p-q-1)$ -sphere.” For this Rambau & Ziegler [451] gave explicit counterexamples: For example, there is a projection of a 5-dimensional polytope (simplicial, 2-neighborly, 10 vertices, 42 facets) to a hexagon such that the poset of all subdivisions is disconnected. Nevertheless, Edelman & Reiner [189] have proved that the generalized Baues conjecture is *true* in the case of a (general position) projection of a simplex into the plane. The important cases of general projections of d -simplices (related to spaces of triangulations) and the projections of d -cubes (related to zonotopal tilings and oriented matroids, see Chapter 7) are still wide open and very tantalizing. (See also Björner [88], Sturmfels [534], and Mnëv & Ziegler [410].)

The figure on the next page shows the poset of all subdivisions for the projection of Example 9.3.

We met a special case of this before: when $P = C_p$ is a p -cube and Q is thus a zonotope, then the set of all zonotopal tilings (those are the tilings of $Z = Q$ by faces of $P = I^p$) is the poset of all one-element oriented matroid extensions, which appears in the Bohne-Dress Theorem 7.32 (Section 7.5).

In contrast to the set of *all* subdivisions, the poset of all π -coherent subdivisions is the face poset of a polytope: of the “fiber polytope” of the projection. In the drawing on the next page, this is the part of the poset drawn with solid lines — the face poset of a hexagon (Exercise 9.1)!

Definition 9.4. Let $\pi : P \rightarrow Q$ be a projection of polytopes. A *section* is a (continuous) map $\gamma : Q \rightarrow P$ that satisfies $\pi \circ \gamma = \text{id}_Q$, that is, $\pi(\gamma(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x} \in Q$.

The *fiber polytope* $\Sigma(P, Q)$ is the set of all average values of the sections of π , that is,

$$\Sigma(P, Q) = \left\{ \frac{1}{\text{vol}(Q)} \int_Q \gamma(\mathbf{x}) d\mathbf{x} : \gamma \text{ is a section of } \pi \right\}.$$

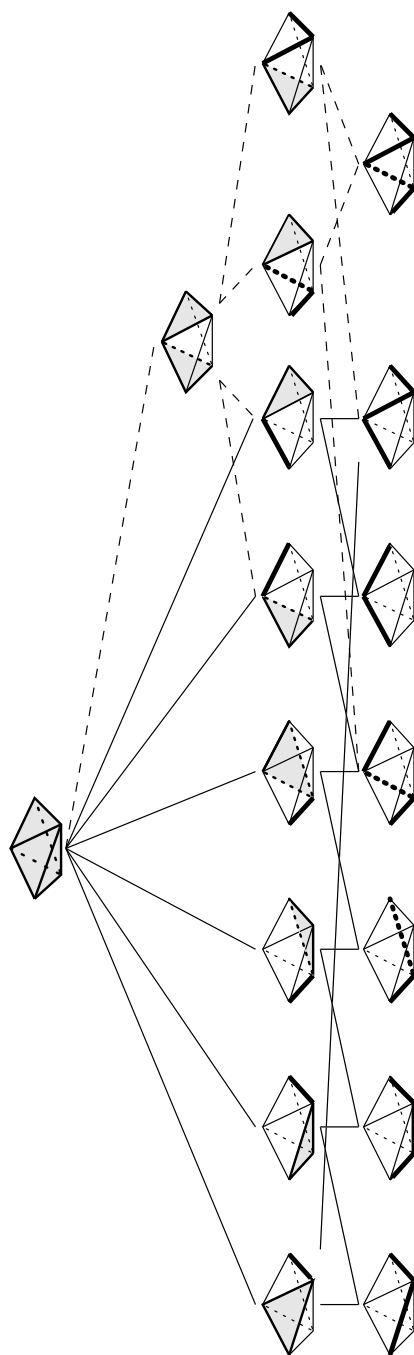
Without loss of generality we can restrict our discussion to those sections that are piecewise linear over a polyhedral subdivision of Q . We can integrate such sections componentwise, using classical Riemann integrals. Here we use that for a linear function f on a polytope R , one has the formula

$$\int_R f(\mathbf{x}) d\mathbf{x} = \text{vol}(R) \cdot f(\mathbf{r}_0),$$

where $\mathbf{r}_0$ is the barycenter of R , $\mathbf{r}_0 = \frac{1}{\text{vol}(R)} \int_R \mathbf{x} d\mathbf{x}$.

Any convex combination of sections is a section as well, and from this we get that the fiber polytope is a convex set. Furthermore, a simple calculation shows that it is contained in the fiber of the barycenter of Q ,

$$\Sigma(P, Q) \subseteq \pi^{-1}(\mathbf{r}_0) \cap P.$$



The scaling factor $1/\text{vol}(Q)$ in the definition is only needed for this inclusion, but is irrelevant for the geometry of the fiber polytope.

Lemma 9.5. *A subdivision given by $\mathcal{F} \subseteq L(P)$ induces a tight π -coherent subdivision of Q if and only if it is a minimal element in the partial order $\omega_{coh}(P, Q)$ of π -coherent subdivisions.*

Proof. From the definition of the partial order it is clear that every tight subdivision is minimal. For the converse, observe that if we have $\mathbf{c} \in (\mathbb{R}^p)^*$ which induces a certain π -coherent subdivision $\mathcal{F}^{\mathbf{c}}$, then we can perturb this $\mathbf{c}$ to general position $\mathbf{c}'$, and get $\mathcal{F}^{\mathbf{c}'}$, and then the resulting subdivision will be tight by construction and smaller (or equal) to the one we started with. $\square$

The following is the key result from Billera & Sturmfels [78].

Theorem 9.6. *$\Sigma(P, Q)$ is a polytope of dimension $\dim(P) - \dim(Q)$, whose nonempty faces correspond to the π -coherent subdivisions of Q , that is, the face lattice of $\Sigma(P, Q)$ is*

$$L(\Sigma(P, Q)) = \{\hat{0}\} \cup \omega_{coh}(P, Q).$$

Here the vertices of $\Sigma(P, Q)$ correspond to the finest π -coherent subdivisions, which are the tight ones, while the facets correspond to the coarsest proper subdivisions.

In particular, in the special case where $P = \Delta_p$ is a simplex we get that the vertices of $\Sigma(\Delta_p, Q)$ correspond to triangulations of Q — this is the case of *secondary polytopes* as considered in [231, 232]; see the next section.

Proof. (Sketch) Any convex combination of two sections is a section again. Linearity of the integral yields from this that the set $\Sigma(P, Q)$ is convex. Its dimension cannot be larger than $\dim(P) - \dim(Q)$, because $\Sigma(P, Q)$ is contained in the fiber $\pi^{-1}(\mathbf{r}_0)$, which has this dimension.

Every piecewise linear section that is not tight can be changed locally in two opposite directions; thus it can be written as a convex combination of two other sections that have a different integral. Furthermore, there are only finitely many different tight sections. Thus we get that the set $\Sigma(P, Q)$ is the convex hull of the integrals $\frac{1}{\text{vol}(Q)} \int_Q \gamma(\mathbf{x}) d\mathbf{x}$ for which γ is a tight (piecewise linear, continuous) section of $\pi : P \rightarrow Q$. From this we conclude that $\Sigma(P, Q)$ is a polytope.

To detect the vertices of $\Sigma(P, Q)$, we use that they arise as the *unique* maxima for *generic* linear functions $\mathbf{c} \in (\mathbb{R}^p)^*$. However, if $\mathbf{c}$ is generic, then every fiber $\pi^{-1}(\mathbf{r})$ for $\mathbf{r} \in Q$ has a unique maximal element with respect to $\mathbf{c}$. This shows that $\mathbf{c}$ determines a unique, tight, coherent section $\gamma^{\mathbf{c}}$ via the construction of Definition 9.2, and the integral over this section is the only point of $\Sigma(P, Q)$ which maximizes $\mathbf{c}$. Thus the vertices of $\Sigma(P, Q)$ coincide with the tight coherent subdivisions $\pi : P \rightarrow Q$.

Every face of $\Sigma(P, Q) \subseteq \mathbb{R}^p$ is defined by some linear function on $\mathbb{R}^p$. Thus it defines a map $\pi^{\mathbf{c}} : P \rightarrow Q^{\mathbf{c}}$, and thus a π -coherent subdivision $\mathcal{F}^{\mathbf{c}}$ of Q . Again a simple computation shows that for a continuous section $\gamma : Q \rightarrow P$, the point $\frac{1}{\text{vol}(Q)} \int \gamma d\mathbf{x} \in \Sigma(P, Q)$ lies in the face defined by $\mathbf{c}$ if and only if the image of the section is entirely contained in the collection of faces $\mathcal{F}^{\mathbf{c}} \subseteq L(P)$.

This yields a bijection between the faces of $\Sigma(P, Q)$ and the coherent subdivisions of Q , and thus in particular, between the *coarsest* such subdivisions and the *facets* of $\Sigma(P, Q)$. $\square$

The correspondence between faces of $\Sigma(P, Q)$ and π -coherent subdivisions of Q also yields an explicit method to construct the vertices and the facets of $\Sigma(P, Q)$, which we will use heavily.

Namely, for every vertex there is a unique section, which we only have to integrate. For the facets we always proceed as follows. For every coarsest subdivision of $Q \subseteq \mathbb{R}^q$ that has a chance to be π -coherent, we construct a “lifting” $\widehat{Q} \subseteq \mathbb{R}^{q+1}$ such that the subdivision comes from the lower faces of $\widehat{Q}$. If we can find the affine map $\pi^{\mathbf{c}} : P \rightarrow \widehat{Q}$, then the function $\mathbf{c}$ that defines the facet can be reconstructed via $\mathbf{c}\mathbf{x} = (\pi^{\mathbf{c}}(\mathbf{x}))_{q+1}$.

9.2 Some Examples

Some special cases of the fiber polytope construction are easily analyzed.

If $Q = \{\mathbf{q}\}$ is a point ($q = 0$), then $\Sigma(P, Q) = \Sigma(P, \{\mathbf{q}\}) = P$.

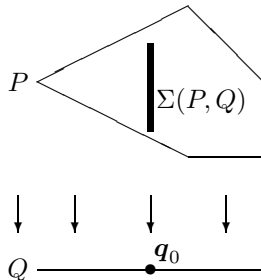
More generally, if $P = R \times Q$ is a product, and $\pi : R \times Q \rightarrow Q$ is the canonical projection, then it is easy to see that $\Sigma(P, Q) = \Sigma(R \times Q, Q) \cong R$: the fiber polytope is a translate of R . In fact, we get

$$\Sigma(R \times Q, Q) = R \times \{\mathbf{q}_0\},$$

where $\mathbf{q}_0$ denotes the barycenter of Q .

If $p = q$, then we have $P = Q$ and $\Sigma(Q, Q) = \{\mathbf{q}_0\}$.

If $p = q + 1$, then $\Sigma(P, Q)$ is an edge $[\mathbf{s}^\downarrow, \mathbf{s}^\uparrow]$, which π maps to the barycenter of Q .



The lower end $\mathbf{s}^\downarrow$ of the interval arises as the integral of the collection $\mathcal{L}^\downarrow$ of “lower faces” of P , while $\mathbf{s}^\uparrow$ is the integral of the collection of “upper faces,” divided by $\text{vol}(Q)$ in both cases.